

Disagreement-Aware Probabilistic U-Net: Making Predictive Diversity Land Where Annotators Actually Disagree

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August 19, 2026

Selected paper. S. A. A. Kohl et al., *A Probabilistic U-Net for Segmentation of Ambiguous Images*, NeurIPS 2018 [1].

Proposed extension. We supervise *where* human annotators disagree. A single-convolution head predicts the inter-rater disagreement map from the image alone, and an alignment loss drives the model’s own sample diversity into those same pixels.

1 Introduction

Semantic segmentation assigns a label to every pixel of an image, and training normally assumes that each image has one correct mask. In medical imaging that assumption does not hold. Several qualified experts asked to outline the same lesion produce visibly different masks, and on the LIDC-IDRI lung-nodule dataset they sometimes disagree about whether a lesion is present at all. A conventional U-Net [4] trained on one expert’s mask, or on the average of several, returns a single confident segmentation and throws that information away precisely where it matters most clinically.

The Probabilistic U-Net addresses this by combining a U-Net with a conditional variational autoencoder. A prior network maps the image x to a distribution $p(z | x)$ over a low-dimensional latent space, and each draw of z is decoded into a different plausible segmentation. Crucially the samples are coherent whole masks rather than independent per-pixel probabilities. This is one of the paper’s main arguments: a pixel-wise probability map cannot express which pixels vary *together*, so it cannot represent two distinct but internally consistent readings of the same image. Training minimises the negative evidence lower bound (ELBO), in which a posterior $q(z | x, Y_g)$ additionally observes one randomly chosen expert mask Y_g ,

$$L_{\text{PU-Net}} = \underbrace{\mathbb{E}_{z \sim q(z|x, Y_g)} [-\log p(Y_g | x, z)]}_{\text{reconstruction (cross-entropy, summed over pixels)}} + \beta \underbrace{D_{\text{KL}}(q(z | x, Y_g) \| p(z | x))}_{\text{pulls the prior towards the posterior}},$$

so that at test time, sampling from the prior alone reproduces the variability seen across annotators.

The resulting distribution of segmentations is scored with the generalized energy distance. Writing S, S' for independent model samples and Y, Y' for independent grader masks,

$$D_{\text{GED}}^2 = 2 \mathbb{E}[d(S, Y)] - \underbrace{\mathbb{E}[d(S, S')]}_{\text{sample diversity}} - \mathbb{E}[d(Y, Y')], \quad d = 1 - \text{IoU},$$

where $\text{IoU}(A, B) = |A \cap B| / |A \cup B|$ is the intersection over union of two binary masks.

We chose this paper because it raised a question we found genuinely interesting: do we always want a model to output the single answer it considers most likely? The paper’s answer is no. When the human experts themselves disagree, there is no single correct mask to predict, and a confident deterministic output hides that. We liked the idea of capturing the disagreement instead of averaging it away, and wanted to take it one step further by asking not only how much the experts disagree but *where*. LIDC-IDRI makes this possible, since it provides four independent expert masks per image, so the disagreement can be measured directly rather than assumed.

2 Proposed extension

The limitation we address is in how the paper’s diversity is supervised and scored. During training the posterior sees one randomly drawn grader per step, so nothing in the objective ever states which *parts* of the image the

graders argued about. The evaluation has the same blind spot: every term of D_{GED}^2 is a single number, with $E[d(S, S')]$ measuring how much the samples vary and $E[d(Y, Y')]$ how much the graders vary, but none of them asks where that variation falls. A model could therefore place the right amount of variability in entirely the wrong region and still score well.

We expected this to be worth fixing for two reasons. First, the information is already in the data and simply unused: four masks per image give a per-pixel picture of the disagreement for free. Second, a model that can point at the disputed region is more useful as a second reader than one that only reports an overall confidence, which is the clinical motivation the paper itself opens with. The change fits the original architecture rather than replacing it, since both of our additions attach to the last U-Net feature map and leave the encoder, decoder, latent space and loss of the original model intact.

Three quantities have to be kept distinct. The first, D^{GT} , is the *human* disagreement, computed from the four masks alone. With $P_i = \frac{1}{G} \sum_g Y_i^{(g)}$ the per-pixel mean over $G = 4$ graders, D^{GT} is its normalised binary entropy,

$$D_i^{\text{GT}} = -\frac{P_i \log(P_i + \epsilon) + (1 - P_i) \log(1 - P_i + \epsilon)}{\log 2} \in [0, 1],$$

which is 0 wherever all four graders agree and 1 where they split two against two. The second, $\hat{D}$, is our model’s *prediction* of D^{GT} . The third, U , is the *model* uncertainty: we draw K latents from the prior, decode each into a foreground-probability map $Q^{(k)}$, average them into $\bar{Q}_i = \frac{1}{K} \sum_k Q_i^{(k)}$ and apply the same normalised binary entropy,

$$U_i = -\frac{\bar{Q}_i \log(\bar{Q}_i + \epsilon) + (1 - \bar{Q}_i) \log(1 - \bar{Q}_i + \epsilon)}{\log 2}.$$

Since U and D^{GT} differ only in what is averaged, model samples in one case and grader votes in the other, any discrepancy between them is a discrepancy of location on a common $[0, 1]$ scale. That is what makes $|U - D^{\text{GT}}|$ a sensible thing to minimise.

Our extension adds two components, both deliberately small. The **disagreement head** is a single 1×1 convolution followed by a sigmoid, applied to the last U-Net feature map and taken *before* the latent z is injected, so that $\hat{D}$ depends on the image alone. It is trained with $L_D = \text{MSE}(\hat{D}, D^{\text{GT}})$. The **alignment loss** $L_{\text{align}} = \frac{1}{HW} \sum_i |U_i - D_i^{\text{GT}}|$ penalises predictive diversity that lands anywhere other than where the graders differ, giving the full objective $L = L_{\text{PU-Net}} + \lambda_D L_D + \lambda_A L_{\text{align}}$. In both cases the idea is to add pixel-level information that answers the “where” question, while keeping the original model’s view of a segmentation as a whole coherent mask. Figure 1 shows both additions in place.

It is worth saying how the two views fit together. D^{GT} and U are per-pixel marginals, that is, projections of the distribution over whole masks that the paper models. We do not replace one with the other. The conditional VAE still produces the distribution, the KL term still keeps the prior broad and the decoded samples coherent, and L_{align} only says where the variance should sit. U itself is computed from the K decoded samples, so it is a readout of that distribution rather than a second, independent pixel-wise model. There is a limit to this. Four masks fix the per-pixel statistics exactly, so with $G = 4$ the map D^{GT} can only take the values $\{0, 0.81, 1\}$, but they say much less about which pixels vary together, so a per-pixel constraint is close to all that four annotations can support.

This gives two testable claims. **Q1:** where annotators will disagree is predictable from the image alone, so $\hat{D}$ should approximate D^{GT} and beat a trivial predictor that only marks lesion boundaries. **Q2:** adding L_{align} should increase the agreement between U and D^{GT} compared with the baseline, without hurting segmentation quality or the energy distance.

We considered two alternatives. Conditioning the prior on disagreement, $p(z \mid x, \hat{D})$, was rejected because $\hat{D}$ is itself a deterministic function of x , so it adds no new information and would blur the comparison between arms. Per-rater output heads or annotator confusion matrices [5] were rejected because the LIDC release does not record which radiologist produced which mask, and the four raters differ from scan to scan, so a per-rater parameterisation would have nothing stable to learn.

3 Methodology

Model. We reimplemented the Probabilistic U-Net in PyTorch, since the official release is an older TensorFlow codebase that no longer runs on Colab. The model has three parts. The U-Net is a standard encoder–decoder with skip connections: the encoder halves the spatial resolution four times while doubling the channel count, starting

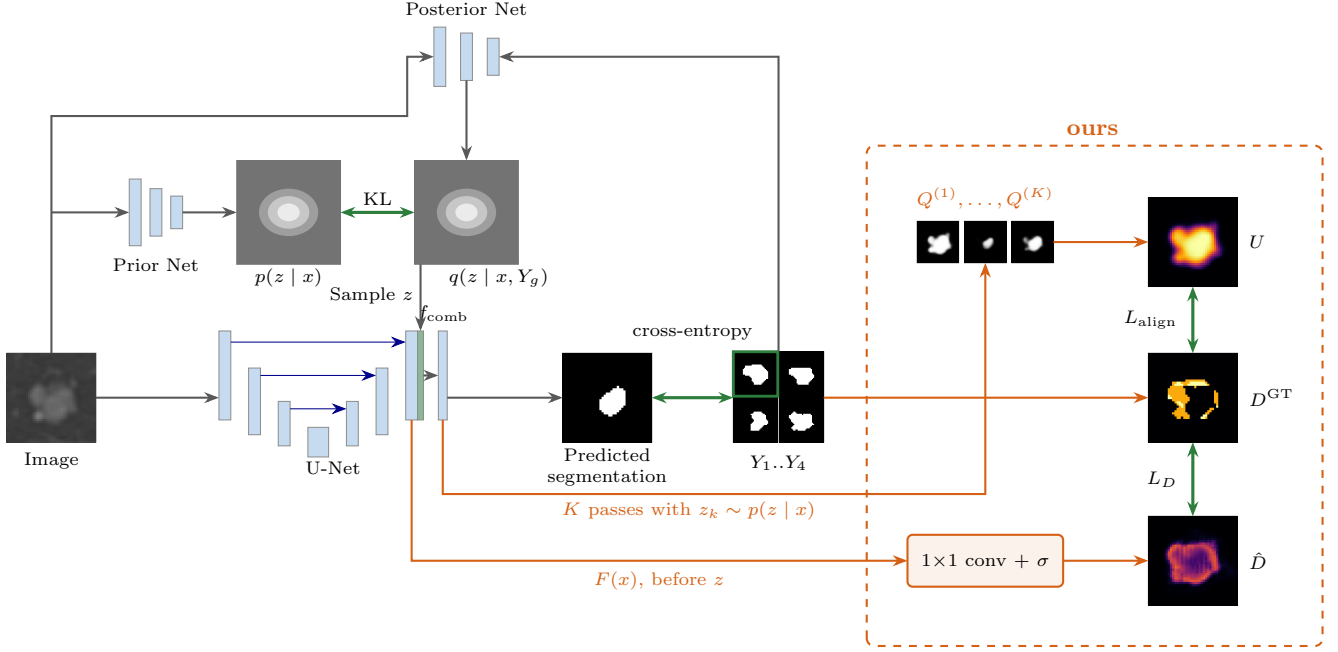


Figure 1: **The disagreement-aware Probabilistic U-Net (training).** Outside the dashed box, the original model: the posterior sees the image and *one* randomly drawn grader Y_g (green outline), and the sampled z (green block, 6 channels) is concatenated to the last U-Net activation (32 channels). The three 1×1 convolutions of f_{comb} combine them. Inside the box, our extension. Note that $F(x)$ is taken *before* z is added, so $\hat{D}$ depends on the image alone, and that the K passes behind U each draw a fresh z_k from the *prior*. **Every thumbnail is real**, from test crop LIDC-IDRI-0217 decoded by our trained full model and zoomed to the nodule; D^{GT} , U and $\hat{D}$ share one $[0, 1]$ colour scale, so they can be compared directly. The three $Q^{(k)}$ shown are non-empty draws: 8 of the 16 samples for this crop are empty, though all four graders segmented the nodule.

from 32 channels, and the decoder mirrors it, concatenating the matching encoder feature map at every step. Alongside it are two small convolutional encoders that each output the mean and variance of a six-dimensional Gaussian. The prior encoder sees only the image; the posterior encoder, used during training, sees the image together with one grader’s mask. A sampled z is broadcast into a six-channel spatial map, concatenated onto the U-Net’s last feature map, and passed through three 1×1 convolutions (the f_{comb}) that mix the two and produce the class logits. Only this last block has to be recomputed to draw another sample, which is what makes sampling cheap. Our disagreement head is a fourth branch: one 1×1 convolution and a sigmoid on that same last feature map, taken before z is concatenated. It lives in a subclass of the baseline model, so all three arms share one implementation.

Data. We use the LIDC-IDRI [3] 2D crops from DeepMind’s preprocessed release (CC BY 3.0), published with the Hierarchical Probabilistic U-Net [2], whose preprocessing follows Appendix H.1 of the original paper. The splits contain 8843 training, 1993 validation and 1980 test images from 530, 111 and 103 patients respectively; images are 180×180 8-bit grayscale with four masks each. We take a random 128×128 crop, scale images to $[0, 1]$ and binarise the masks. These counts are up to 0.6% below the 8882 / 1996 / 1992 reported in the paper, because the authors cleaned the dataset after publication and the 2018 snapshot was never released. About 65% of training crops contain at least one empty mask, which reflects disagreement about whether a lesion exists at all rather than about where its boundary lies. This turns out to matter for how the metrics are computed.

Training. The objective, optimiser and schedule are taken unchanged from the original paper: the reconstruction cross-entropy summed over pixels, $\beta = 1$ with a six-dimensional latent, Adam with learning rate 10^{-4} decayed to 10^{-6} in five steps, batch size 32 and weight decay 10^{-5} . We did not tune any of these, and every arm uses them identically, which is what lets the results be read as an ablation rather than as the outcome of a search.

Three choices were ours. We train for 100k steps instead of the paper’s 240k, which was a matter of available compute. Our absolute numbers are therefore not directly comparable with the published ones, but since all three

arms were trained for the same number of steps, comparing them with each other is still fair. We set $\lambda_D = 0.01$ and $\lambda_A = 10^{-5}$ after short 10k-step trials over a range of values, which showed that larger weights on the auxiliary terms buy very little and cost a great deal: at $\lambda_A = 10^{-3}$ the uncertainty correlation improved by roughly 0.02 while Dice fell from 0.30 to 0.12. We kept the smallest weights that still had a measurable effect. Finally, we clip the gradient norm at 100 and clamp the latent log-variance, both needed to keep training stable. Augmentation is limited to the random translated crop, where the paper also uses elastic deformation, rotation, shearing and scaling. Training draws $K = 4$ samples per step and evaluation uses 16 samples per image.

Ablation design. Three arms vary one component at a time.

Arm	Disagreement head	Alignment loss	Objective
A – baseline PU-Net	no	no	$L_{\text{PU-Net}}$
B – + head	yes	no	$+\lambda_D L_D$
C – full model	yes	yes	$+\lambda_D L_D + \lambda_A L_{\text{align}}$

Tools. Python, PyTorch, NumPy and Matplotlib. The submitted code runs in Google Colab as a self-contained notebook that executes the whole pipeline in a reduced configuration for a quick demonstration; the same code, run at full length, produced every number reported below.

4 Results and analysis

Metrics. Dice and IoU, with $\text{Dice}(A, B) = 2|A \cap B|/(|A| + |B|)$, check that the extension does not damage segmentation quality. We compute them between the model’s deterministic prediction and each of the four masks in turn. They are not our headline claim. The generalized energy distance compares our 16 samples with the four human masks. Because both IoU and Dice are 0/0 when the two masks being compared are empty, some convention is needed for those pairs. For the energy distance we follow the paper’s Appendix B, which sets $d \equiv 0$, so that agreeing there is no lesion counts as agreement. The paper does not say what to do for Dice, and there we made our own choice: we drop empty-versus-empty pairs from the average rather than scoring them zero. Scoring them zero would penalise every model equally on the 65% of crops that contain an empty mask and would hide the differences we care about. The correlation $\text{corr}(U, D^{\text{GT}})$, which asks whether the model is uncertain in the same places the humans disagree, is our primary result.

Two further diagnostics are needed. Disagreement in LIDC concentrates on lesion boundaries, so a map that merely marks a boundary already correlates with D^{GT} . We therefore compare $\hat{D}$ against a trivial control: a 5-pixel band around the edge of the model’s own predicted mask, reaching about two pixels inside and two outside. The control uses no grader information at all, so unlike a band drawn around the ground truth it is something a deployed model could actually compute. We also report the $E[d(S, S')]$ term of the energy distance on its own, since it detects samples that have collapsed onto a single mask, and *nestedness*: the fraction of images for which, after sorting that image’s samples by area, each one is wholly contained in the next. Nestedness catches a subtler failure, where the model only inflates and shrinks one shape without ever proposing a different one, so its samples can look varied while really having a single degree of freedom.

Main results. Table 1 reports the held-out test split (1980 images, 16 samples per image), with every arm read from its 100k-step checkpoint.

Arm	Dice $\uparrow$	IoU $\uparrow$	GED $\downarrow$	$\text{corr}(U, D^{\text{GT}}) \uparrow$	$\text{corr}(\hat{D}, D^{\text{GT}}) \uparrow$	control	$E[d(S, S')]$	nested $\downarrow$
A – baseline	0.355	0.280	0.394	0.483	—	0.417	0.686	1%
B – + head	0.308	0.241	0.403	0.492	0.609	0.451	0.687	3%
C – full	0.340	0.267	0.388	0.509	0.627	0.425	0.692	1%

Table 1: Test-set results. *control* is the boundary-band predictor for the same model, so $\text{corr}(\hat{D}, D^{\text{GT}})$ should be read against the value beside it.

Q1 holds. The head reaches $\text{corr}(\hat{D}, D^{\text{GT}}) = 0.627$ against its own boundary-band control of 0.425, a margin of +0.20, and the same pattern appears in both arms that carry a head. The head has therefore learned something

about how the annotators behave, not merely retraced the outline of its own prediction. This is the clearest positive result we obtained.

Q2 is supported, but the effect is small. The correlation $\text{corr}(U, D^{\text{GT}})$ rises from 0.483 to 0.509 between arms A and C, and arm C at the same time reaches the best energy distance of the three (0.388) for a 4% relative drop in Dice. We think the modest size of the gain is structural rather than a matter of tuning. L_{align} can only see the per-pixel marginal, so two sample distributions that share a marginal look identical to it: the loss can move variance around, but it cannot change which masks the model proposes.

Getting even this far required fixing a bug in our own implementation. Our `fcomb` originally applied no nonlinearity between its 1×1 convolutions, which made the whole block a single affine map. Since z enters as a spatially constant vector, every sample was then a level set of one fixed function, and 100% of test images had strictly nested samples: z could only grow or shrink one mask, never move part of a boundary and leave the rest. With the activations applied, nestedness drops to the 1–3% in Table 1, and only then does the alignment loss have anything to act on.

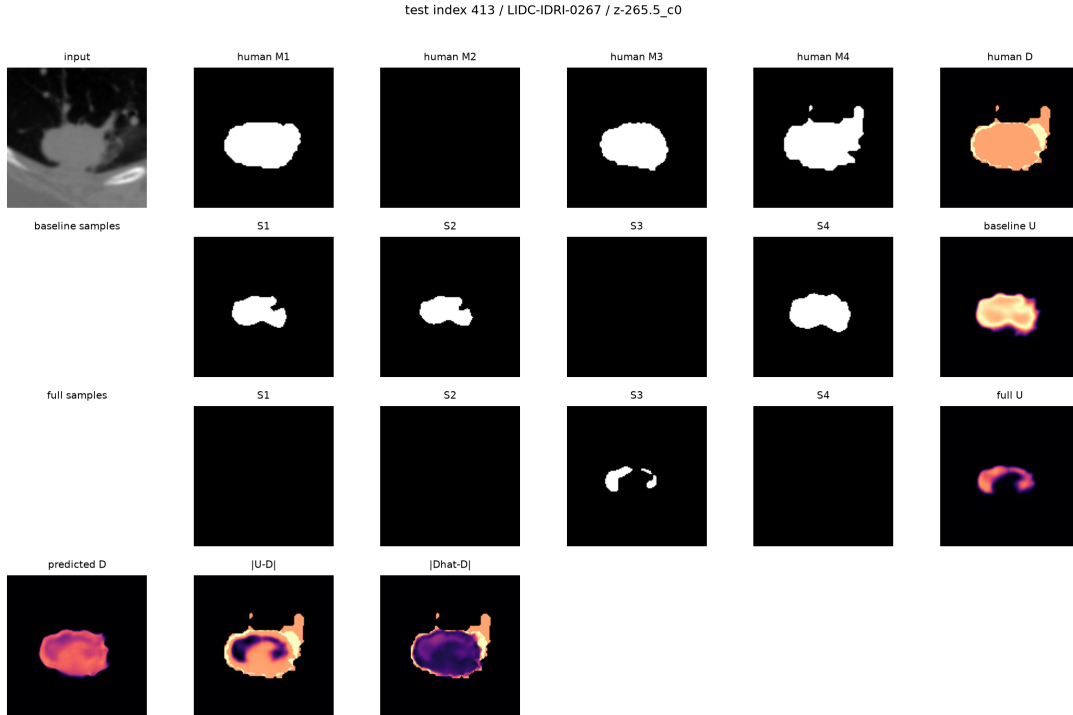


Figure 2: A case where the graders disagree about whether a lesion exists at all (test index 413, LIDC-IDRI-0267). Row 1 shows the CT crop, the four expert masks, of which M2 is empty, and D^{GT} , which covers the whole nodule rather than a rim because the disagreement is about existence rather than boundary placement. Row 2 shows four segmentations sampled from the baseline and the uncertainty map U they produce; row 3 shows the same for our full model. Both reproduce the ambiguity, in that some of their samples contain the nodule and some are empty. Row 4 shows the predicted $\hat{D}$ and the two error maps. $|\hat{D} - D^{\text{GT}}|$ is dark over the lesion while $|U - D^{\text{GT}}|$ is bright: the head recovers this kind of disagreement, but U would have to be maximal across the entire nodule to match D^{GT} , which pulls directly against the segmentation loss.

Challenges. The most instructive problem was a posterior collapse in our first full-length run, which we discarded. We had averaged the reconstruction cross-entropy over pixels where the original paper [1] sums it. At 128×128 that under-weights the reconstruction by a factor of about 16000 against β KL, so $\beta = 1$ behaved like $\beta \approx 16000$. The KL term reached zero by step 31k and the validation energy distance rose from 0.34 to 0.61 while Dice kept improving, which is the signature of a network that has quietly become deterministic. Summing the cross-entropy fixed it, and it was a useful reminder that in a VAE the reduction used for the reconstruction term is not a formatting detail.

Checkpoint selection turned out to be misleading in a similar way. Choosing the checkpoint with the best

validation energy distance picks step 1000 for arm A and step 3000 for arm B, because a barely-trained model with a wide, uninformative prior scores well on that metric. We therefore report all arms at the same 100k steps, which compares the methods rather than how long they happened to train.

Finally, Figure 2 points at a limit of the idea itself. Much of the disagreement in LIDC is about whether a nodule is there at all, so asking U to match D^{GT} asks the model to be maximally uncertain across whole lesions, which works directly against the reconstruction term. This bounds how far L_{align} can usefully be pushed.

5 Conclusion

We set out to make a Probabilistic U-Net say not just how much its samples should vary but where. Two findings came out of it. Where radiologists will disagree is predictable from the image alone: a single 1×1 convolution reaches a correlation of 0.627 with the true disagreement map, against 0.425 for a trivial boundary-band predictor. Aligning the sample diversity with that disagreement gives a smaller but consistent gain, raising $\text{corr}(U, D^{\text{GT}})$ by 0.026 while reaching the best energy distance of our three arms at a 4% Dice cost. The claim we want to make is not that our Dice is higher; the baseline already produces diverse plausible segmentations. It is that our version puts that diversity where the experts actually disagree, and makes the disagreement itself an output one can look at.

The main limitation is that D^{GT} and U are per-pixel quantities. Two graders drawing one mask and two drawing a clearly different one can produce the same D^{GT} as four graders disagreeing pixel by pixel, so neither map distinguishes two coherent interpretations from unstructured noise. This bounds both what we can supervise and what we can measure. Beyond that, we trained for 100k of the paper’s 240k iterations and used only the random crop for augmentation, so our absolute energy distances sit above the published ones; we tested one dataset and one anatomy; D^{GT} is estimated from just four graders; and each arm was run once, with a single seed, so the smaller differences between arms, including arm B’s lower Dice, cannot be separated from run-to-run variation.

Two directions look worth taking. Restricting L_{align} to a band around the predicted boundary would let it shape where edges are drawn without demanding uncertainty across whole lesions, which we think is the most likely route to a clearer Q2 result. It would also be informative to compare against the Hierarchical Probabilistic U-Net, a follow-up by the same authors that models ambiguity at several spatial scales instead of through one global latent, and which we would expect to handle existence-versus-boundary disagreement better than our single six-dimensional z .

References

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